

Data Science

Grid Search and Pipelines

Machine Learning Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Implement scikit-learn pipelines to create reproducible machine learning workflows.

2

Apply **GridSearchCV** to systematically search for optimal hyperparameters.

3

Evaluate and interpret grid search results to select the best model configuration.

Tuning Hyperparameters



Trivia

What is a hyperparameter?



Trivia

What is a hyperparameter?

A hyperparameter is a model configuration setting chosen before training begins, rather than learned from data during training.

Training Machine Learning Models

As you begin working with more complex machine learning algorithms, you will have more and more control over how the models run. For example, k in k -nearest neighbors.

Selecting these **hyperparameters** can be challenging and time consuming.... libraries such as [scikit-learn](#) have tools to help with this process.



Grid Search

What Is Grid Search?

Grid search is a systematic approach to hyperparameter tuning.

- Exhaustively evaluates all possible combinations of parameter values specified in a grid.
- Rather than manually testing different configurations, grid search automates this process to find the **optimal set of hyperparameters** for your model.



Grid Search

Let's see it for ourselves!

- Open **02-gridsearch-pipelines.ipynb**



Practical Grid Search Use

The search space grows exponentially with each additional parameter.

Requires careful selection of parameter ranges to be computationally feasible.

Consider RandomizedSearchCV for high-dimensional parameter spaces.

Always set appropriate evaluation metrics aligned with your problem.

Pipelines

Pipelines

Pipelines in **scikit-learn** provide a way to chain multiple processing steps into a **single, unified workflow**.

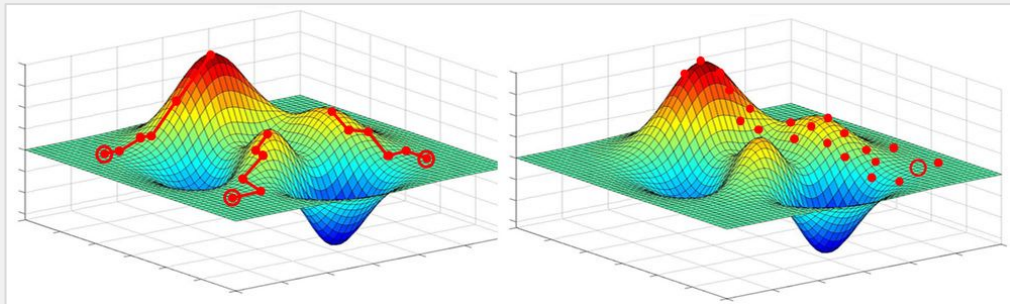
This can reduce errors and ensure consistent data preprocessing every single time you load new data.



Pipelines

Pipelines facilitate the model building process by:

- Speeding up hyperparameter tuning.
 - ◆ Manual hyperparameter tuning is time-consuming.
- Preventing errors in the preprocessing steps.
 - ◆ Such as data leakage.
- Provide a systematic approach to model selection and evaluation.



Pipelines

Let's see it for ourselves!

- Open **02-gridsearch-pipelines.ipynb**



Practical Pipeline Use

- Pay attention to Memory management.
 - ◆ Large datasets may cause memory issues when transformations create copies; use memory parameter to cache intermediate results.
- Errors can cause major issues.
 - ◆ Failures in early pipeline steps affect all subsequent steps.
 - ◆ Implement robust error handling.
- Debugging can be more challenging in pipelines.
 - ◆ Errors within pipeline steps can be harder to troubleshoot; test individual components before combining.

Apply Pipelines

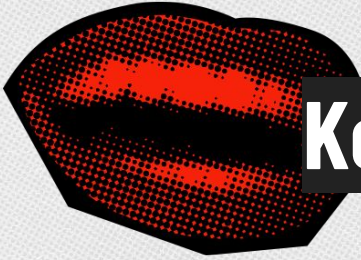
Pipelines

Let's see it for ourselves!

- Open **gridsearch-pipelines-exercise-starter-code.ipynb**



Bring It Home



Key takeaways



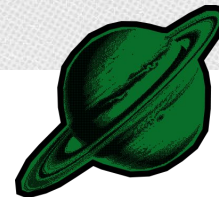
Super Charge your Workflow

Grid search and pipelines create reliable reproducible machine learning workflows.



Error Reduction

When used carefully you can reduce errors with pipelines.



Foundation for Advanced ML!

Pipelines are critical for more advanced ML such as ensemble methods and NLP.

